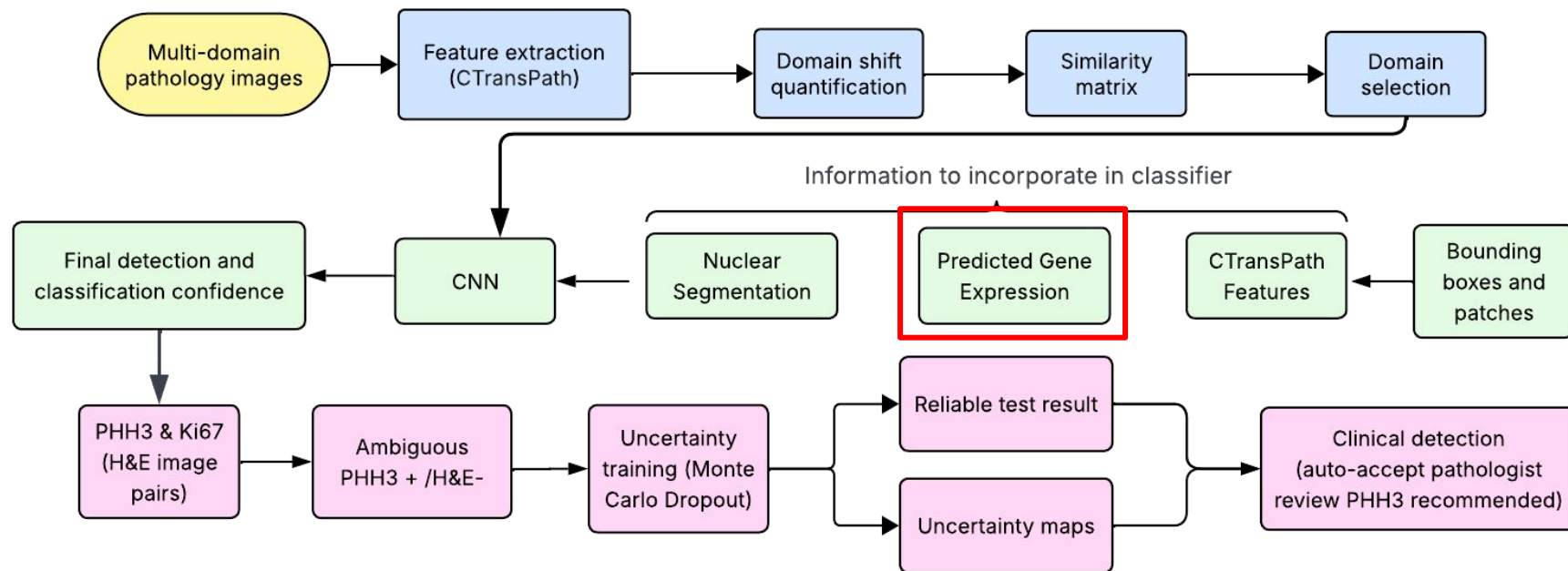




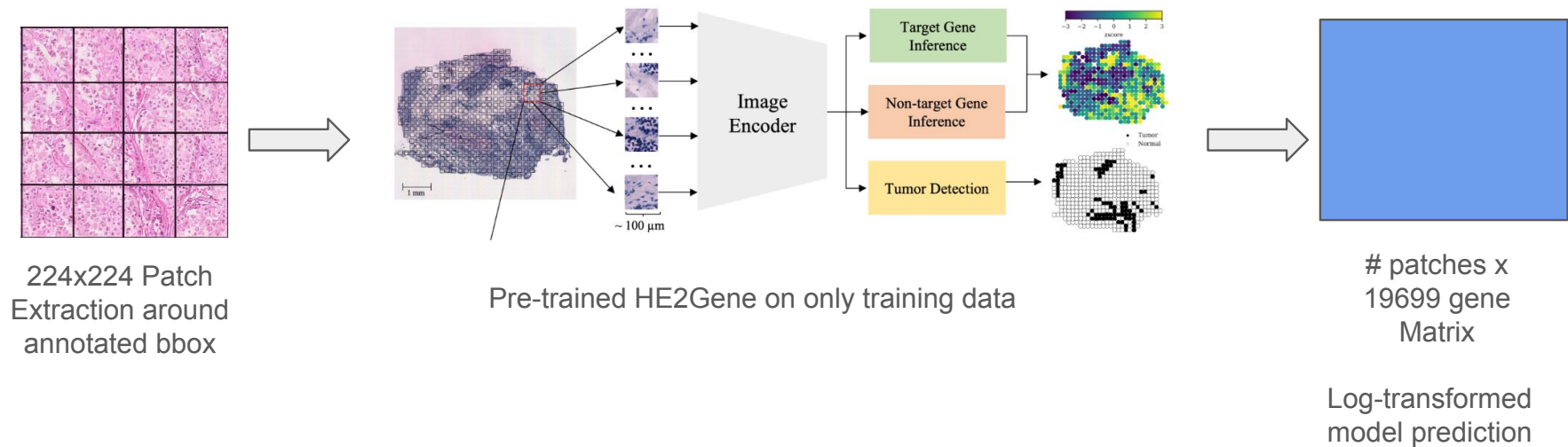
2/12 Update Presentation

Lillian Lam | Naina Misra | Yang Cao | 02/12/2025

Pipeline



HE2Gene



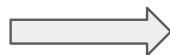
From Github:

“Generalizes well on an external dataset without the need for re-training”

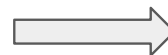
Differential Expression workflow

Transform into
AnnData Object

21650 patches x
19699 genes

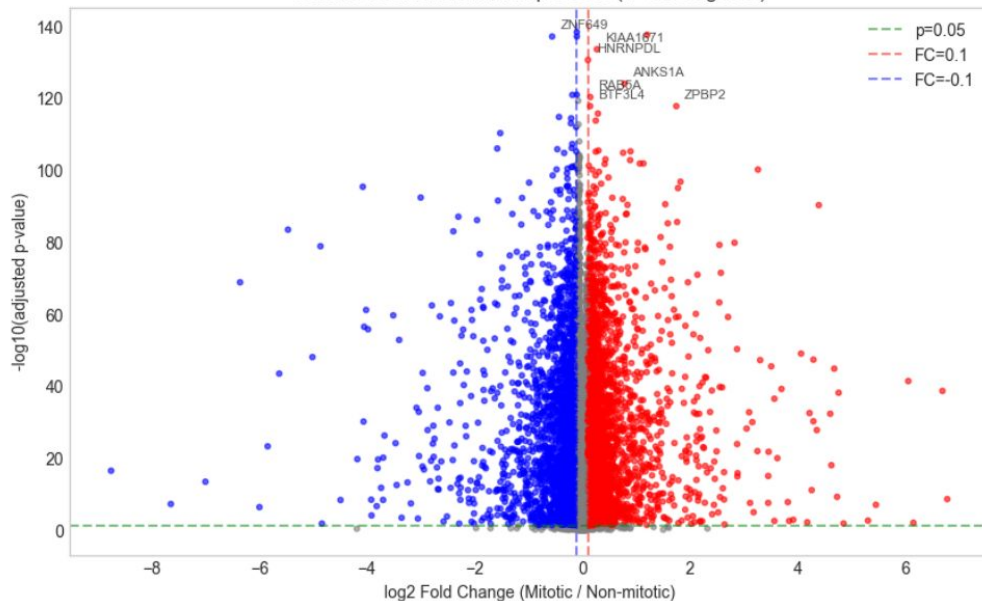


Rank Genes Group
(Wilcox-test &
benjamini-hochberg
correction)

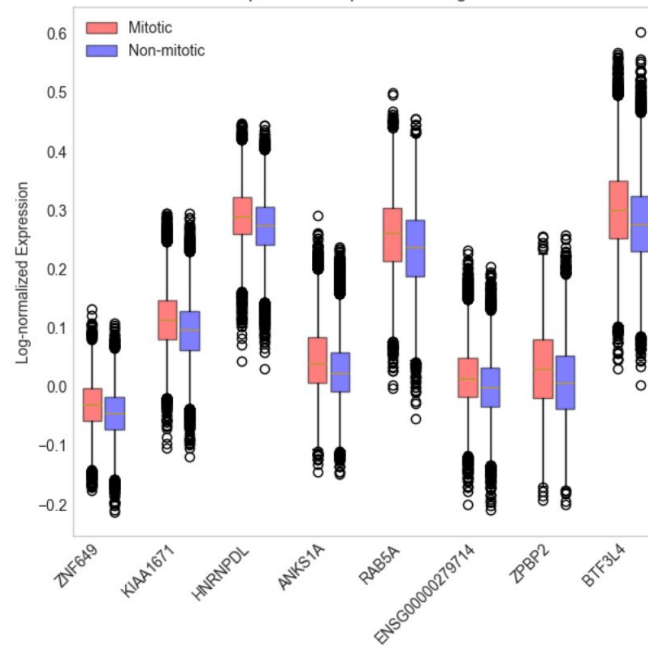


Significant Genes
Identified
(adj.p < 0.05 &
log2FC > 0.1)

Volcano Plot: Differential Expression (n=19699 genes)



Top 8 Genes Up in Mitotic Figures

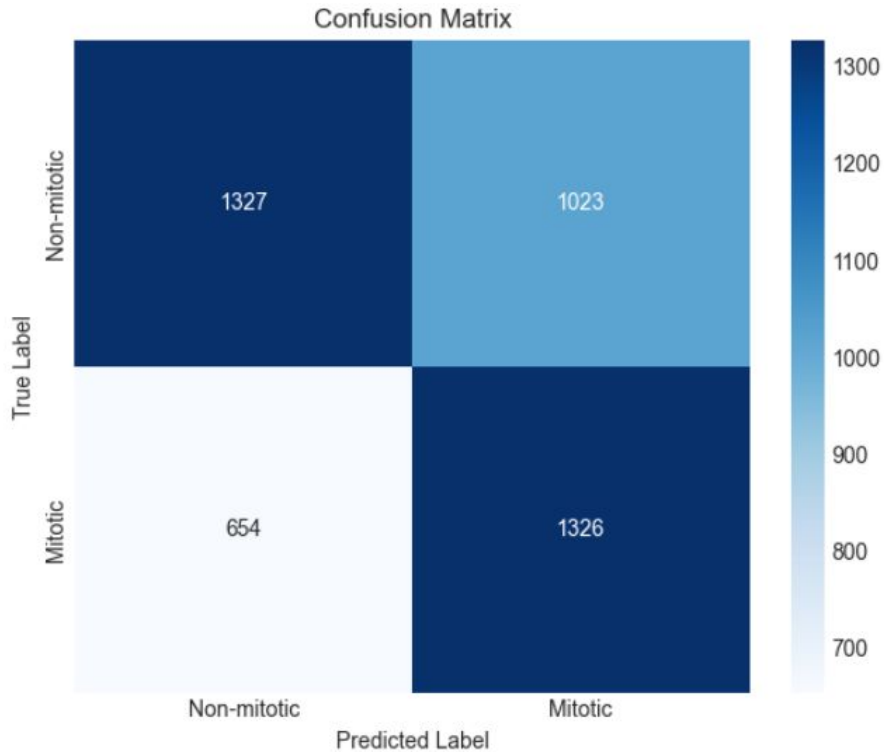


Training

Metric	Value	Interpretation
Training samples	17,320	80% of total patches
Test samples	4,330	20% of total patches
Features	100 genes	Top DE genes by adjusted p-value
Classifier	SVM (RBF kernel)	class_weight='balanced'

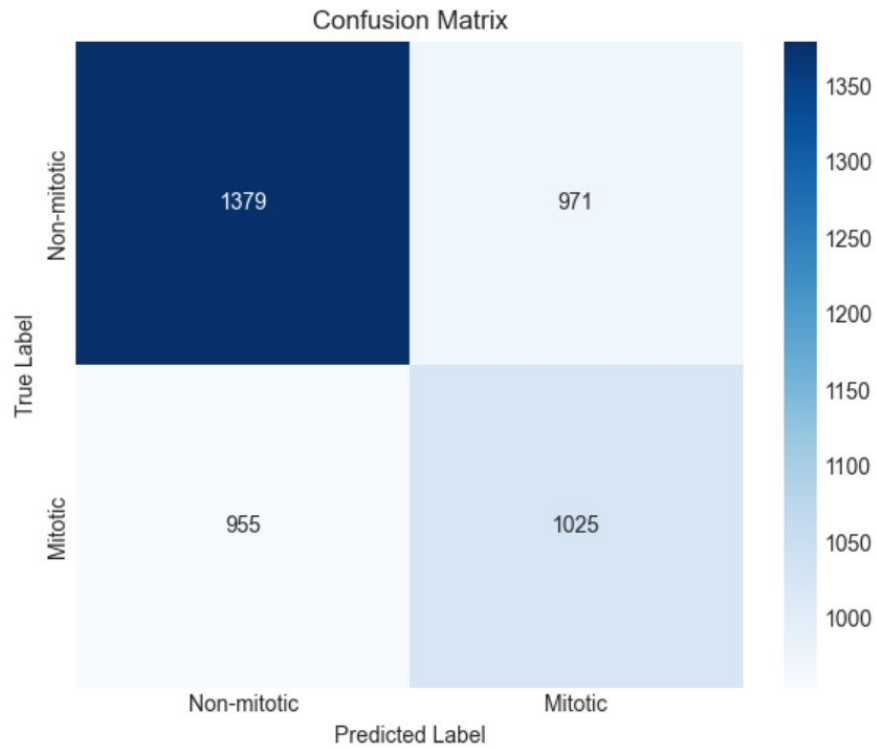
Results

Metric	Cross-Validation (5-fold)	Test Set
Accuracy	0.608 ± 0.008	0.613
F1-Score	0.607 ± 0.004	0.613
ROC-AUC	0.653 ± 0.008	0.666
Precision		0.564
Recall		0.67



Results with known mitotic markers

Metric	Cross-Validation (5-fold)	Test Set
Accuracy	0.565 $\pm$ 0.009	0.555
F1-Score	0.531 $\pm$ 0.009	0.516
ROC-AUC	0.585 $\pm$ 0.011	0.586
Precision		0.514
Recall		0.518



Summary

- AURKA, PLK1, CCNB1, CDK1, MKI67 do NOT discriminate
 - known mitotic figures fail
- Discovered signatures do not overlap with experimentally validated mitotic markers
 - ZNF649, ARAP1 are likely model artifacts

Next Time

- Started training with R-CNN with ResNet-50 backbone
- Rank similarity of H&E WSI images to incorporate into training

Ideas

- Try other H&E to ST models (STimage, HiST, etc.)
- Incorporate only Hematoxylin in the training (nucleic acid staining)
- Use CTransPath as the backbone for R-CNN
- Train on Ki67 datasets to learn proliferation morphology
 - o train on subset of Ki67 positive to distinguish mitotic figures

Thank you!